

**Yoink!**

Cambridge University Library claims that a thief may have stolen 2 of Charles Darwin's notebooks. <https://digit.in/dec20-35>

**Automated decipherment**

Researchers have built a system that can translate extinct ancient languages, using ML. <https://digit.in/dec20-36>

Translating unknown languages

New technologies could help scientist unravel the mysteries of ancient lost languages

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ne of the frustrating things for archaeologists, is to come across inscriptions that they cannot read.

All around the world, there are various examples of lost scripts, languages that we simply cannot understand. We do not understand Linear Elamite, a language that originated in what is now Iran, the writing systems of the Minoans and the Cretans, the inscriptions on the Phaistos Disc, and several Mesoamerican scripts including the Zapotec and Olmec writing systems. Closer back home the script used by the people in the Indus Valley Civilisation remains a mystery, and has evaded several decades of attempts to decipher it. It becomes easier for humans to decipher a lost language, if there is an example of text in the lost language, and a translation in a known language. The Rosetta Stone discovered in 1799 when Napoleon led his soldiers into Egypt, had an inscription in Egyptian hieroglyphics, another Egyptian script called Demotic and Greek. It was this stone that allowed linguists to decipher the Egyptian scripts. Even Google Translate works



The symbols on the Phaistos disc

Credit: Ozgur Gonen / Shutterstock.com

on the same principle, only with many many different sets of translations between inscriptions.

However, many of the lost languages mentioned above do not have their rosetta stones, and there are no translations in a known language. Guessing the meaning of an unknown script from scratch was a painstaking process for humans, and considered impossible for computers. Back in 2010, MIT professor Regina Barzilay and colleagues set out to use computers to decipher unknown scripts.

The system they built works on a few assumptions, and considers a language that is known to be closely related to the unknown language. This would allow for a way to map the letters of the unknown alphabet to the letters of a known alphabet. In both the languages, certain clusters of glyphs would tend to appear together, because of the relationship between the two languages. The relation also

implies that the languages have some cognate words, that share the same root and have a similar meaning, pronunciation and spelling. For example, the Spanish and Latin words for “controversy” are “controversia” and “controversus”. Then the system also assumes that certain words will be mapped in a similar way, and have similar structures. For example, the word overloading has both a prefix and a suffix, “over” and “ing”, both of which can be found in other words, such as “oversee” and “finding”. At times, both the suffix and prefix can appear in another word, such as “overcrowding”. This allows the scientists to guess that the word that means “overloading” in another language related to English, will also have a three part structure with a suffix and a prefix. The French word for “overloading”, is “surchargeant”.

The system works by iterating the data hundreds and even thou-